

Place Value, Rounding and Estimating — Paper D

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to come out exactly, or to be estimated.
- The rules for powers of ten, rounding and significant figures are not printed on this paper. You are expected to know them.
- When you estimate, round every number to 1 significant figure and show what you rounded each one to — that is usually the first mark.
- Several questions ask whether an estimate is too big or too small. Track what each rounding did to the answer; if the two roundings pull in opposite directions, saying so is a correct answer.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – CHAINS, AND WHICH NUMBERS ROUND WHERE

1. Work out:

(a) $0.8 \times 0.1 \times 0.01$ [1]

(b) $4 \div 0.01 \div 0.1$ [1]

(c) $25 \times 0.01 \div 0.1$ [1]

(d) $0.06 \div 0.1 \times 0.1$ [1]

2. A whole number is rounded to the nearest 100 and the answer is 500.

(a) Write down the smallest whole number this could have been. [1]

(b) Write down the largest whole number this could have been. [1]

(c) How many whole numbers round to 500 to the nearest 100? Show how you counted them. [2]

SECTION 2 – ACCURACY, AND WHAT IT HIDES

3. Round 0.0298471 to:

(a) 2 decimal places [1]

(b) 4 decimal places [1]

(c) 2 significant figures [1]

(d) 3 significant figures [1]

4. A machine produces 0.68 kg of powder every minute. It runs for 4950 minutes.

- (a)** Estimate the total mass produced, by rounding each number to 1 significant figure. [2]

- (b)** Work out the exact total mass. [1]

- (c)** State whether your estimate is bigger or smaller than the exact answer, and explain why. [1]

- 5.** A mass is recorded as 3.60 kg, correct to 2 decimal places.

- (a)** Write down the smallest value the true mass could be. [1]

- (b)** Describe carefully the largest value the true mass could be. [1]

- (c)** Explain why writing 3.6 kg instead of 3.60 kg would give the reader different information. [2]

- 6.** Two students were each asked to round 5.7449 to 2 decimal places. [5]

Student V wrote: “5.7449 → 5.745 → 5.75”

Student W wrote: “the 9 on the end makes it round up, so 5.75”

For each student, describe exactly what they did wrong. Then give the correct answer.

- 7.** A student has written two answers in his book, and both are wrong. [5]

(i) “ $6.2 \div 0.01 = 0.062$, because dividing makes numbers smaller.”

(ii) “0.04728 to 2 significant figures is 0.05.”

For each one, explain the mistake and give the correct answer. Then state, for each, the general rule the student has missed.

End of paper. Check every answer has a unit or a form the question actually asked for.